

Maintenance sheet version 10 for accumulation area (single mast, two levels)

Station name	SDM
People present	APA, JJ, BAK, ARU, Eggert, Björn
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	23/5 - 2024

Notes: Upgrade from GC-Net AWS v3 to GC-Net AWS v4 (vaisala)

Issue: new logs w/ old modem not responding ~15:30 (departure)
 changed to new modem → worked
 30053406437800
 same modem transferred to new logger box

AWS to do list		Check after maintenance			
Part	Shipped? (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	Y	Y	212887	110723	Clean? Y (N) rime both top & bottom
Iridium modem IMEI number	N	N	5154530	5154530	→ clean at departure
Inclinometer/compass (Inside CNR4)	Y	Y	Attached to the radiometer? Y/N		
Satellite antenna	N	N			
Wind sensor upper	Y	Y	Black box pointing toward the radiometer (i.e., along the boom)? Y/N		
Wind sensor lower	Y	Y	Black box pointing same direction as the radiometer? Y/N		
Temperature / humidity assembly upper	N (lufft) Y (vaisala)	Y	Fan spins? Y/N Sounds, OK? Y/N		
Temperature / humidity assembly lower	N (lufft) Y (vaisala)	Y	Fan spins? Y/N Sounds, OK? Y/N		
Sonic ranger upper	Y	Y	Old sensor with new membrane? Y/N		
Sonic ranger lower	Y	Y	Old sensor with new membrane? Y/N		
Thermistor string	Y	Y	New system		
Solar panel	N	N	Clean? Y/N NB: is not effective when partially covered		
GPS antenna	N Y	Y	At top of logger enclosure? Y/N		
Logger box ID	Y	M4-06 Y	CR1000 or CR1000X? ID number: M4-06		
Logger enclosure	Y	Y	Replaced (including everything inside)? Y/N New desiccant bags? Y/N		
Battery box	N	N	New box? Y/N New desiccant bags? Y/N		
Mast extension inserted?	Y	Y	How many inserted? How long? Notes: 1 150m standard		
Mast	N	N	New mast? Y/N New sensor boom? Y/N		

battery box flush with surface 70 cm
 west of mast underneath lower Vaisala

GNSS mounted 2cm below the mast connection (see picture)

APL started ~15:15 LT, passed firn core ~15:32 LT

260 + 174

Station name	SDM 73, RPA, BAV, ARU, Egypt, B. Ym
Date	23/5-2024

Metadata maintenance

Arrival: -24°C, low wind, 10cm snow

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	$\pm$ 00:68	00:00
Download data to PC and/or change SD card (wait for green light)	☒ Y / N	Y / N
Name of logger program	GC-Net 2.4	GC-Net 3.2 OS 0602
Screen dump of values in fast scan mode	☒ Y / N	☒ Y / N
Photos of mast, sensors, logger box wiring etc. (more is better)	☒ Y / N	Y / N
Mast tilt towards radiometer (- if looking up)	0°	°
Mast tilt across boom looking from radiometer (- if clockwise)	+1°	°
Upper boom direction with radiometer relative to north	° true/magn.	° true/magn.
Lower boom perpendicular to upper boom. If not, measure.	☒ Y / N	☒ Y / N
Box on upper wind sensor exactly same direction as radiometer? If not, measure.	☒ Y / N	☒ Y / N
Box on lower wind sensor exactly same direction as radiometer? If not, measure.	☒ Y / N	☒ Y / N
Upper temperature / humidity height (bottom of casing - surface)	275 cm	295 cm
Lower temperature / humidity height (bottom of casing - surface)	165 cm	434 cm
Upper sonic ranger height (bottom of casing - surface)	240 cm	393 cm
Lower sonic ranger height (bottom of casing - surface)	124 cm	279 cm
Length of thermistor string on surface from surface marking	— cm	— cm
Apparent damage to sensors? If yes, which?	Y / N	Y / N

Notes: Probe 160 cm to battery box lid
lower boom height above surface 138 cm
rim on SW down up dome

snowboard depth: 180 cm (probing 10m from mast)
sb stake below snow surface

→ Upper Luft not tightened, instrument slid down to its casing resting on infinity bolt
new SB, stake carbon 158 + 74 thin bamboo extension out of snow
loggerbox ID: M3-0002

Must-do at next visit:

two mast extensions (1.8 m snowfall)

two x 2m carbon stake for snowboard

bring 1 new Luft for precip!

→ None of the Lufts were working (on AWS at arrival)

Thermistor: 159 + 71 cm out of snow (to surface)

↑
plastic tube

↑
thin bamboo extension